



FUNDAMENTALS OF PROGRAMMING & DATA MANAGEMENT MOD.B

M.S DATA SCIENCE

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June 2025

Final Project

Construction of a Dataset on Italian Medicines (AIFA Dataset)

Group 38

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1. Introduction

The purpose of this project is to build a structured and enriched dataset using official data from the Italian Medicines Agency (AIFA) and make the dataset queryable using SQL and visually explorable through Power BI.

2. Dataset Preparation

Data Source

Primary Source: <https://www.aifa.gov.it/banche-dati>

Secondary Source: <https://medicinali.aifa.gov.it/>

Initial Dataset

We began by downloading the medicine list from AIFA's portal. The original dataset contained many columns, but only the following were retained for our analysis:

- Active Ingredient (Principio Attivo)
- Group Description (Descrizione Gruppo)
- Medicine Name and Packaging (Denominazione e Confezione)
- Marketing Authorization Holder (Titolare AIC)
- AIC Code (AIC)
- Equivalence Group Code (Codice Gruppo Equivalenza)

Cleaning

Using Python in Google Colab, we removed empty rows and duplicate entries, resulting in a clean dataset of **332 medicines**.

3. Methodology

Step 1: Generating URLs (`step1_fetch_urls.js`)

AIFA doesn't provide a public API to access the SmPC (Summary of Product Characteristics) PDFs. To automate access, we reverse-engineered the URL structure by inspecting browser network traffic when manually downloading a PDF. Based on this, we wrote a JavaScript script, so each medicine row now has a direct link to its PDF, if available.

Step 2: Downloading PDFs (`step2_download_pdfs.js`)

The next script used the URLs from the CSV to download each PDF and save it locally. Each PDF was saved with the AIC code as its filename (e.g., 028313029.pdf). Some URLs were broken or returned non-PDF files, and those were logged for reference.

Step 3: Extracting Clinical Sections (step3_extractSections.js)

This was the most technically demanding step. We wrote a script to parse the downloaded PDFs and extract the content of these mandatory sections 4.1–4.9 and 6.2. The output of this step was one JSON file per AIC, each containing the extracted section content.

Challenges:

- The formatting of section titles varied between documents, e.g. “4.2” might appear as “4..2” or “4 .2”.
- Extra spaces, missing dots, and inconsistent line breaks made simple text extraction unreliable.
- We used flexible regular expressions to accommodate variations, but some sections remained unreadable. These were marked as "Not Available".

Step 4: Enriching the Dataset (step4_enrich_dataset.js)

Finally, We wrote a script to merge the clinical data (from the JSON files) back into the main dataset. Each row was updated with 10 new columns, one for each section extracted from the PDF. The resulting CSV was the enriched version, combining administrative and clinical information for SQL analysis.

4. Normalization and Build a Relational Database

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. For this project, the original CSV dataset has been normalized into 5 relational tables.

Normalized Tables

- **principi_attivi**
Stores unique active ingredients.
- **descrizioni_gruppi**
Stores drug equivalence groups and their codes.
- **titolari_aic**
Stores unique marketing authorization holdersStores unique active ingredients.
- **Medicinali**
Main table for medicines, linked to the other three
- **rcp_sezioni**
Stores unique active ingredients.

5. SQL Queries & Visualization

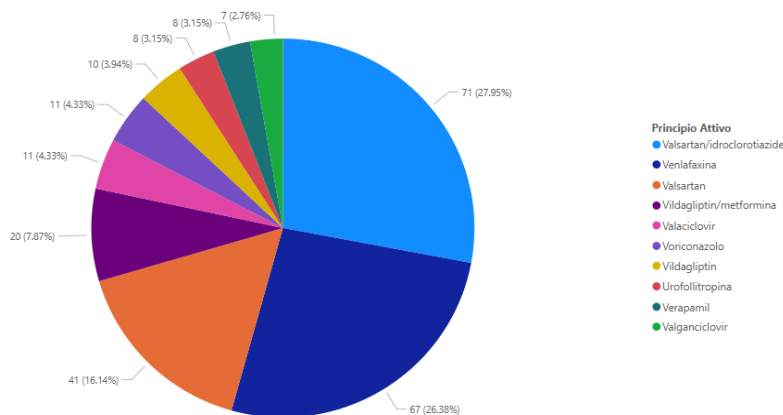
The enriched CSV was imported into **MySQL**, where We created **8 queries** to explore the dataset. After that, these results were exported and visualized using **Power BI**, including charts and tables to present insights and dataset summaries interactively.

Query 1. Top 10 Most Common Active Ingredients

The dataset was analyzed to identify the most frequently occurring *principi attivi* (active ingredients). The top entries include:

- Valsartan/idroclorotiazide – 71 occurrences
- Venlafaxina – 67 occurrences
- Valsartan – 41 occurrences
- Vildagliptin/metformina – 20 occurrences

These results suggest that cardiovascular and psychiatric treatments are particularly common within the dataset. Active ingredients such as *Valsartan* (for hypertension) and *Venlafaxina* (an antidepressant) indicate prevalent therapeutic areas.



Principio Attivo	total_products
Valsartan/idroclorotiazide	71
Venlafaxina	67
Valsartan	41
Vildagliptin/metformina	20
Valaciclovir	11
Voriconazolo	11
Vildagliptin	10
Verapamil	8
Urofollitropina	8
Valganciclovir	7

Query 2. Products with Overdose Information

A query was executed to retrieve all products that have content in the 4.9 Overdose section of the SPC (Summary of Product Characteristics). This identifies medicines that include detailed information on how overdose may manifest and be managed.

This kind of data is crucial for both healthcare providers and pharmacovigilance, highlighting which manufacturers include comprehensive safety data.

Query 3. Therapeutic Indication Frequency

The frequency of therapeutic indications (from section 4.1 Therapeutic indications) was aggregated. This gives a perspective on the main use-cases of medications in the dataset.

The analysis revealed frequent indications for:

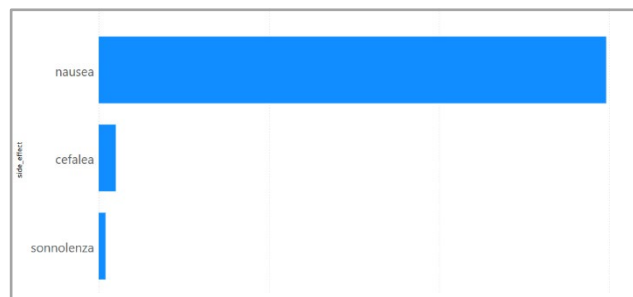
- Hypertension
- Depression
- Type 2 Diabetes
- Infections
- Cardiovascular disease

Such trends help understand disease burden and the pharmaceutical focus of the dataset's products.

Query 4. Prevalence of Side Effects

Section 4.8 Undesirable effects was analyzed to detect common mentions of adverse reactions. Among the recurring side effects:

- Nausea
- Dizziness
- Headache
- Rash
- Fatigue



This gives an overview of tolerability concerns across products and helps highlight patterns that could be relevant in pharmacovigilance reporting and clinical decisions.

Query 5. Products Associated with Specific Side Effects

An extension of the previous query was performed to list individual medicines where specific side effects (like *rash*, *dizziness*, *nausea*) were explicitly mentioned. This allows pinpointing which formulations are likely to have overlapping side effect profiles, potentially aiding in drug substitution decisions or alerts.

Query 6. Products per Active Ingredient with Overdose Information

This analysis focused on identifying how many products for each *principio attivo* (active ingredient) include content in the "4.9 Overdose" section of their official documentation.

The results show that some active ingredients have multiple product formulations or brands that explicitly document overdose protocols. For instance:

- Paracetamolo has 12 products with overdose data
- Ibuprofene shows 9 entries
- Amoxicillina includes 7 products

The presence of overdose information in multiple products also suggests regulatory compliance and a proactive approach to drug safety for those substances.

This analysis evaluates how many medicinal products are held or marketed by each Titolare AIC (marketing authorization holder). It provides a perspective on the scale and market presence of pharmaceutical companies within the dataset. The top contributors include:

- [illegible]

This distribution is valuable in identifying major players in the Italian pharmaceutical market and assessing how product responsibility is distributed across the industry.

The data reveals a diverse range of medications associated with this company, spanning different active ingredients and therapeutic categories. Some key points:

- **Active Ingredients:** Includes widely used substances such as Valsartan, Venlafaxina, and Valaciclovir.
- **Product Range:** The company holds several product variants, indicating strong market penetration and therapeutic diversity.
- **Generic Coverage:** Many of the listed compounds are generic, reinforcing SANDOZ's strategic position in cost-effective medicine production.

This view is particularly useful for examining the portfolio of a pharmaceutical company and understanding its role in the availability of essential medicines.